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Cloud connectivity options for Amazon Web Services (AWS)

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Background

Data centre and cloud architectures are increasingly bespoke. As new technologies and environments arise, so do new tools from cloud providers. Today's digital architects have an overwhelming number of solutions for each use case they encounter, and that includes connectivity options.

The right cloud connectivity option depends on the use case and connectivity goals. It may also depend on the business goals, roadmap for cloud adoption, urgency to modernise and staffing expertise.

Purpose

The purpose of this document is to outline the Amazon Web Services (AWS) Virtual Private Cloud (VPC) connectivity options from on-premises to the cloud environment, factoring in the government Secure by Design approach. We also look at the advantages and limitations of each option.

More about Secure by Design

- [Guide to adopting Secure by Design](#)
- [Secure by Design policy](#)
- [Implementing Secure by Design](#)

1. Choosing a cloud connectivity solution

When choosing a cloud connectivity option in AWS, there are several factors to consider to ensure you select the best solution for your organisation's requirements.

1.1 Use case and application requirements

Latency sensitivity: Real-time applications (for example, VoIP or video streaming) require low latency connections.

Bandwidth requirements: Data-intensive workloads like backups, analytics or media processing may need high bandwidth throughput.

Availability and reliability: Critical applications may need failover links to ensure highly available connectivity.

1.2 Security requirements

Encryption: VPN encrypts traffic over the internet. AWS Direct Connect can be paired with MAC Security (MACsec) or VPN encryption.

Compliance: Consider the compliance requirements, including data privacy and data loss prevention (DLP) controls.

1.3 Cost considerations

Setup and ongoing costs: A VPN is low cost and easy to set up.

AWS Direct Connect involves set-up fees and ongoing port/data transfer costs.

Data transfer pricing: Analyse AWS egress charges and compare with VPN and AWS Direct Connect.

1.4 Geographic reach and AWS region availability

Ensure the connectivity option is supported in your desired AWS region.

Consider proximity to Direct Connect locations or edge locations for better performance.

1.5 Scalability and growth

VPNs may not scale well with large organisations.

AWS Direct Connect + AWS Transit Gateway is better for large, multi-VPC or hybrid environments.

1.6 Failover and redundancy

Consider backup connectivity (for example, using VPN as a backup to AWS Direct Connect).

AWS supports Border Gateway Protocol (BGP) failover for high availability.

1.7 Management and operational overhead

A VPN is simpler to manage but may require ongoing monitoring for performance.

AWS Direct Connect requires coordination with a network provider and physical infrastructure.

1.8 Hybrid cloud and multi-cloud needs

If you're integrating AWS with other clouds (for example, Google Cloud Platform (GCP) or Azure), consider third-party connectivity providers or software-defined wide area network (SD-WAN) solutions.

2. Cloud connectivity options

Amazon VPC provides multiple network connectivity options for you to use, depending on your current network designs and requirements. These connectivity options include using either the internet or an AWS Direct Connect connection as the network backbone and terminating the connection into AWS or user-managed network endpoints.

Additionally, with AWS, you can choose how network routing is delivered between Amazon VPC and your networks, leveraging either AWS or user-managed network equipment and routes.

From a Secure by Design perspective, we recommend you implement the applicable activities after your gap analysis exercise, especially:

- [Sourcing a threat assessment](#)
- [Performing threat modelling](#)
- [Performing a security risk assessment](#)

Perform threat modelling, using the Secure by Design guidance to:

- understand the weak points in the selected connectivity option
- develop and implement mitigation tasks

2.1 Network-to-Amazon VPC connectivity options

This section provides design patterns for connecting remote networks with your Amazon VPC environment. These options are useful for integrating AWS resources with your existing on-site services (for example, monitoring, authentication, security, data or other systems) by extending your internal networks into the AWS Cloud. This network extension also allows your internal users to seamlessly connect to resources hosted on AWS just like any other internally facing resource.

VPC connectivity to remote customer networks is best achieved when using non-overlapping IP ranges for each network being connected. For example, if you'd like to connect one or more VPCs to your corporate network, make sure they are configured with unique Classless Inter-Domain Routing (CIDR) ranges. AWS recommends allocating a single, contiguous, non-overlapping CIDR block to be used by each VPC.

2.1.1 AWS Site-to-Site VPN

AWS Site-to-Site VPN is the most cost-effective option for creating connectivity from on-premises to AWS cloud. You can configure your existing firewall, either software or hardware, on the on-premises network as a customer gateway device. The device can now be configured to work with a site-to-site VPN connection to AWS through the console.

Its advantages are:

- you can reuse existing VPN equipment and processes
- you can reuse existing internet connections
- it offers an AWS-managed high-availability VPN service
- it supports static routes or dynamic BGP peering and routing policies

Its limitations are:

- network latency, variability and availability are dependent on internet conditions
- you are responsible for implementing redundancy and failover

It's important to note that a site-to-site configuration is generally used to connect to one VPC.

2.1.2 AWS Transit Gateway + AWS Site-to-Site VPN

This approach is optimal for a scalable VPN to connect multiple VPCs and customer edge devices on-premises and to create VPN IPSEC tunnels. You can use AWS native tools to configure Transit Gateway's VPN attachment. On the AWS end, create a VPN attachment through AWS's VPC service portal. The benefit of this configuration is that it allows native attachments to act as a hub to multiple VPCs.

Its advantages are the same as those of AWS Site-to-Site VPN as well as offering an AWS-managed high availability and scalability regional network hub for up to 5,000 attachments.

One of its limitations is that network latency, availability is dependent on internet connection.

It's important to note that native tunnels like these are designed for basic connectivity use cases, so advanced features such as policy match, IP address masking, traffic inspection, advanced firewall analysis and so on are not supported.

2.1.3 AWS Direct Connect

AWS Direct Connect is a dedicated, private fibre connection to AWS. It allows you to connect an on-premise router to an AWS Direct Connect router over 1 gigabit or 10 gigabit ethernet fibre optic cables. However, some have opted for hosted connections to support smaller bandwidths (that is, 50Mbps, 100Mbps, 200Mbps, 300Mbps, 400Mbps, 500Mbps, 1Gbps, 2Gbps, 5Gbps), which can be ordered through approved AWS Direct Connect Partners.

Direct Connect is optimal for private connectivity to Amazon VPC, EC2, S3 and DynamoDB using virtual interfaces, bypassing internet service providers in your network path. It provides more consistent network performance and lower cost data-out rates compared to the internet.

This option helps to establish a private connection from your remote network to Amazon VPC.

Its advantages are that it offers:

- more predictable network performance
- reduced bandwidth costs
- support for BGP peering and routing policies

One limitation is that it might require additional telecom and hosting provider relationships or new network circuits to be provisioned.

You should note that:

- Direct Connect is a single, non-redundant connection. For redundancy, consider dual Direct Connect connections or an AWS-managed VPN option
- the maximum number of links in a link aggregation group (LAG) is 4, for dedicated 1Gbps and 10Gbps connections only

2.1.4 AWS Direct Connect + AWS Transit Gateway

AWS Direct Connect + AWS Transit Gateway, uses a transit virtual interface (VIF) attachment to a Direct Connect gateway to enable your network to connect several regional centralised routers over a private dedicated connection.

This option helps to establish a private, logical connection from your remote network to a regional network hub for Amazon VPCs, using AWS Direct Connect and AWS Transit Gateway.

One advantage is that it offers an AWS-managed high availability and scalability regional network hub for up to 5,000 attachments.

One limitation is that it might require additional telecom and hosting provider relationships or new network circuits to be provisioned.

2.1.5 AWS Direct Connect + AWS Site-to-Site VPN

With AWS Direct Connect + AWS Site-to-Site VPN, you can combine AWS Direct Connect connections with an AWS-managed VPN solution. AWS Direct Connect public VIFs establish a dedicated network connection between your network and public AWS resources such as an AWS Site-to-Site VPN endpoint. Once you establish the connection to the service, you can create IPsec connections to the corresponding Amazon VPC virtual private gateways.

This option helps to establish a private, encrypted connection from your remote network to Amazon VPC.

Its advantages include:

- reduced bandwidth costs
- more predictable network performance
- support for BGP peering and routing policies on AWS Direct Connect
- being able to reuse existing VPN equipment and processes
- AWS-managed high availability VPN service support for static routes or dynamic Border Gateway Protocol (BGP) peering and routing policies on VPN connection

Its limitations are that:

- it may require additional telecom and hosting provider relationships or new network circuits to be provisioned
- you are responsible for implementing redundancy and failover
- remote device must support single-hop BGP (when leveraging BGP for dynamic routing)

2.1.6 AWS Direct Connect + AWS Transit Gateway + AWS Site-to-Site VPN

This option helps to establish a private, encrypted connection from your remote network to a regional network hub for Amazon VPCs, using AWS Direct Connect, AWS Transit Gateway and AWS Site-to-Site VPN.

With this setup, you can enable end-to-end IPsec-encrypted connections between your networks and a regional centralised router for Amazon VPCs over a private dedicated connection.

You can use AWS Direct Connect public VIFs to first establish a dedicated network connection between your network to public AWS resources, such as AWS Site-to-Site VPN endpoints. Once this connection is established, you can create an IPsec connection to AWS Transit Gateway.

Its advantages are the same as those of AWS Direct Connect + AWS Site-to-Site VPN as well as offering an AWS-managed high availability and scalability regional network hub for up to 5,000 attachments.

Its limitations are:

- it may require additional telecom and hosting provider relationships or new network circuits to be provisioned
- you are responsible for implementing redundancy and failover
- remote devices must support single-hop BGP (when leveraging BGP for dynamic routing)

2.1.7 AWS VPN CloudHub

By using this setup you can securely communicate from one site to another using the AWS VPN CloudHub. The AWS VPN CloudHub operates on a simple hub-and-spoke model that you can use with or without a VPC. Use this approach if you have multiple branch offices and existing internet connections and would like to implement a convenient, potentially low-cost hub-and-spoke model for primary or backup connectivity between these remote offices.

This option helps to establish a hub-and-spoke model for connecting remote offices.

Its advantages are that it:

- allows you to reuse existing internet connections and AWS VPN connections
- offers an AWS-managed high availability VPN service
- supports BGP for exchanging routes and routing priorities

Its limitations are:

- network latency, variability and availability are dependent on the internet
- user-managed branch office endpoints are responsible for implementing redundancy and failover

2.1.8 Software VPN

This option helps to establish a VPN connection from your equipment on a remote network to a user-managed software VPN appliance running inside an Amazon VPC.

Amazon VPC offers you the flexibility to fully manage both sides of your Amazon VPC connectivity by creating a VPN connection between your remote network and a software VPN appliance running in your Amazon VPC network.

This option is recommended if you must manage both ends of the VPN connection, either for compliance purposes or for leveraging gateway devices that are not currently supported by Amazon VPC's VPN solution.

Its advantages are that it:

- supports a wider array of VPN vendors, products, and protocols
- offers a fully customer-managed solution

One limitation is that if required, you are responsible for implementing high availability solutions for all VPN endpoints.

2.1.9 AWS Transit Gateway + SD-WAN solutions

SD-WANs are used to connect your data centres, offices or colocation environments over different transit networks (such as the public internet, MPLS networks, or the AWS backbone using AWS Direct Connect). They manage the traffic automatically and dynamically across the most appropriate and efficient path based on network conditions, application type or quality of service (QoS) requirements.

This option helps to establish the integration of SD-WAN solutions to interconnect several remote locations to a regional network hub for Amazon VPCs, using the AWS backbone or the internet as a transit network.

Its advantages are that:

- it supports a wider array of SD-WAN vendors, products, and protocols

- some vendors offer integration with AWS native services

One limitation is that you are responsible for implementing high availability of the SD-WAN appliances if they are placed in an Amazon VPC.

2.2 Amazon VPC-to-Amazon VPC connectivity options

Use these design patterns when you want to integrate multiple Amazon VPCs into a larger virtual network. This is useful if you require multiple VPCs due to security, billing, presence in multiple regions, or internal charge-back requirements, to more easily integrate AWS resources between Amazon VPCs. You can also combine these patterns with the Network-to-Amazon VPC connectivity options for creating a corporate network that spans remote networks and multiple VPCs.

VPC connectivity between VPCs is best achieved when using non-overlapping IP ranges for each VPC being connected. For example, if you'd like to connect multiple VPCs, make sure each VPC is configured with unique Classless Inter-Domain Routing (CIDR) ranges. Therefore, we advise you to allocate a single, contiguous, non-overlapping CIDR block to be used by each VPC.

2.2.1 VPC peering

A VPC peering connection is a networking connection between 2 VPCs that enables routing using each VPC's private IP addresses as if they were in the same network. VPC peering connections can be created between your own VPCs or with a VPC in another AWS account. VPC peering also supports inter-region peering. Traffic using inter-region VPC Peering always stays on the global AWS backbone and never traverses the public internet, thereby reducing threat vectors, such as common exploits and DDoS attacks.

One advantage is that it leverages AWS-managed scalable networking infrastructure.

Its limitations are that:

- VPC peering does not support transitive peering relationships
- it can be difficult to manage at scale

2.2.2 AWS Transit Gateway

As workloads grow on AWS, scaling networks across multiple accounts and VPCs can incur management and cost overhead, even if VPC peering is used. Transit Gateway is a service that allows you to connect multiple VPCs with on-premises networks to a single connection from a central gateway. It can act as a hub that controls how traffic is routed across connected networks which act like spokes.

AWS users leverage Transit Gateway for simplified management and reduced operational costs, since each network must only connect to Transit Gateway, as opposed to every other network. Scaling is critical in enterprise accounts, as each new VPC connected to the Transit Gateway can be automatically available to every other connected network. It's also important to note that Transit Gateway supports multicast and inter-regional peering. This option helps to connect Amazon VPCs within and across regions using AWS Transit Gateway in a hub-and-spoke model.

Its advantages are that it offers:

- an AWS-managed high availability and scalability service
- a regional network hub for up to 5,000 attachments

One limitation is that Transit Gateway peering only supports static routes.

2.2.3 AWS PrivateLink

AWS PrivateLink enables private connectivity between VPCs, on-premises applications and other AWS services securely on the Amazon network. It offers a simplified private connectivity between accounts and VPCs under an organisation without configuring firewall rules, path definitions or route tables. The private connectivity function negates a requirement for an Internet Gateway or VPC peering connection.

Users can leverage services from another VPC within the AWS network in a highly scalable, one-to-many topology. Since multiple endpoint services can be accessed, PrivateLink can benefit from the elastic load balancing and hourly endpoint cost model.

This option helps to connect Amazon VPCs with VPC interface endpoints and VPC endpoint services.

One advantage is that it leverages AWS managed scalable networking infrastructure.

A limitation is that VPC Endpoint services are only available in the AWS region in which they are created.

2.2.4 Software VPN

Amazon VPC provides network routing flexibility. This includes the ability to create secure VPN tunnels between 2 or more software VPN appliances to connect multiple VPCs into a larger virtual private network. This means that instances in each VPC can seamlessly connect to each other using private IP addresses.

This option is recommended when you want to manage both ends of the VPN connection using your preferred VPN software provider. It uses an internet gateway attached to each VPC to facilitate communication between the software VPN appliances.

This option helps to connect Amazon VPCs using VPN connections established between user-managed software VPN appliances running inside of each Amazon VPC.

Its advantages are that:

- it supports a wide array of VPN vendors, products, and protocols
- it's managed entirely by you

Its limitations are that:

- you are responsible for implementing HA solutions for all VPN endpoints (if required)
- VPN instances could become a network bottleneck

2.2.5 Software VPN-to-AWS Site-to-Site VPN

This option helps to connect Amazon VPCs with a VPN connection established between a user-managed software VPN appliance in one Amazon VPC and AWS Site-to-Site VPN attached to the other Amazon VPC.

Amazon VPC provides the flexibility to combine the AWS-managed VPN and software VPN options to connect multiple VPCs. With this design, you can create secure VPN tunnels between a software VPN appliance and a virtual private gateway, allowing instances in each VPC to seamlessly connect to each other using private IP addresses. This option uses a virtual private gateway in one Amazon VPC and a combination of an internet gateway and software VPN appliance in another Amazon VPC.

Its advantages are that it:

- offers an AWS-managed high availability VPC VPN connection
- supports a wide array of VPN vendors and products managed by you
- supports static routes and dynamic BGP peering and routing policies

Its limitations are that:

- you are responsible for implementing high availability solutions for the software appliance VPN endpoints (if required)
- VPN instances could become a network

2.3 Software remote access-to-Amazon VPC connectivity options

With software remote access VPN, you can leverage low cost, elastic and secure services to implement remote-access solutions while also providing a seamless experience connecting to AWS hosted resources. This option is typically preferred by smaller companies with less extensive remote networks or who have not already built and deployed remote access solutions for their employees.

You can combine these patterns with the Network-to-Amazon VPC connectivity options and Amazon VPC-to-Amazon VPC connectivity options to create a network that spans remote networks and multiple VPCs.

2.3.1 AWS Client VPN

AWS Client VPN is an AWS-managed high availability and scalability service enabling secure software remote access. It provides the option of creating a secure Transport Layer Security (TLS) connection between remote clients and your Amazon VPCs to securely access AWS resources and on-premises over the internet.

This option helps to connect software remote access to Amazon VPC, leveraging AWS Client VPN.

One advantage is that it offers an AWS-managed high availability and scalability service. One limitation is that it's only available to OpenVPN clients.

2.3.2 Software client VPN

You can choose from an ecosystem of multiple partners and open source communities that have produced remote-access solutions that run on Amazon EC2 (Elastic Compute Cloud). These solutions provide great flexibility on the security protocol used for remote access into your Amazon VPCs, to securely access AWS resources and on-premises over the internet.

This option helps to connect software remote access to Amazon VPC, leveraging user-managed software VPN appliances.

Its advantages are that it:

- Supports a wider array of VPN vendors, products and protocols
- Offers a fully customer-managed solution

One limitation is that you are responsible for implementing high availability solutions.

2.4 Transit VPC

A transit VPC is a common strategy for connecting multiple, geographically dispersed VPCs and remote networks in order to create a global network transit centre. A transit VPC simplifies network management and minimises the number of connections required to connect multiple VPCs and remote networks.

This option helps to establish a global transit network on AWS using a software VPN in conjunction with an AWS-managed VPN.

2.5 AWS Cloud WAN

AWS Cloud WAN is an intent-driven managed wide area network (WAN), described by a policy you define that unifies your data centre, branch and AWS networks.

While you can create your own global network by interconnecting multiple Transit Gateways across regions, Cloud WAN provides built-in automation, segmentation and configuration management features designed specifically for building and operating global networks, based on your core network policy. Cloud WAN has added features such as automated VPC attachments, integrated performance monitoring and centralised configuration.

The core network policy is written in a declarative language that defines segments, AWS region routing and how the attachments should map to segments. With a core network policy, you can describe your intent for access control and traffic routing, while AWS Cloud WAN handles the network configuration details.

Cloud WAN is managed within AWS Network Manager, which enables you to centrally manage and visualise your Cloud WAN core network and Transit Gateway networks across AWS accounts, regions and on-premises locations. Network Manager provides you several dashboard visualisations to help you view and monitor all aspects of your global network.

2.6 Summary of AWS connectivity options

Clearly, there are a significant number of factors to consider when choosing a connectivity approach.

We recommend starting by identifying your use case, understanding the current landscape and prioritising goals.

After that, identify requirements of the most critical goals and align them with organisational spend.

Finally, develop a phased approach with rollback plans.

Service	Limits
AWS Native VPN	• Supports up to 1.25Gbps network throughput per VPN tunnel • 4 links for Link Aggregation Groups (LAGs) • Advertise up to 100 routes
AWS Site-to-Site VPN	Supports up to 50 Customer Gateways and 50 Site-to-Site VPN connections per region
AWS Transit Gateway with VPN	Maximum 5 transit gateway attachments per VPC
AWS Direct Connect	Advertise up to 100 routes over each Border Gateway Protocol session
AWS Direct Connect Gateway	• Connect up to 5,000 VPCs and on-premises environments through a single gateway • Multi-account support allows associating up to 10 VPCs or up to 3 AWS Transit Gateways from multiple AWS accounts
AWS Transit Gateway	Regional scalability as a network hub for up to 5,000 attachments
AWS PrivateLink	Default bandwidth for each interface endpoint supports up to 10Gbps per Availability Zone, bursting up to 40Gbps

2.7 More AWS resources

For a detailed overview of any cloud connectivity options mentioned in this guidance, visit the [AWS documentation website](#).